

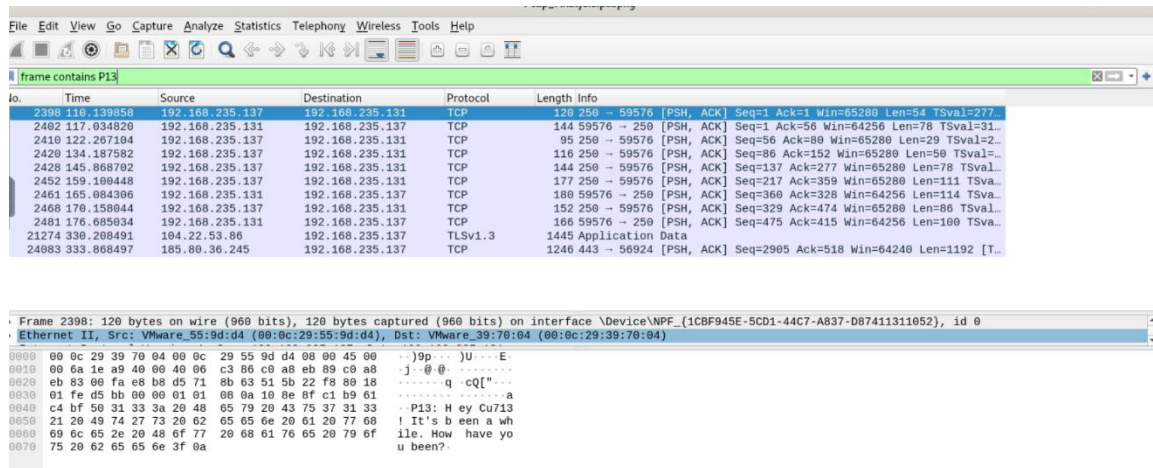
PCAP Analysis – [Write Up]

We have captured this traffic from P13's computer. Can you help him?

| Start Investigation:

Q1: In network communication, what are the IP addresses of the sender and receiver?

Initially, I noticed that there were several repeated protocols, such as UDP and TLS, so I couldn't find what I was looking for. I then focused on the TSP protocol. Also, as explained in the lab, the device was P13, so I searched for any packet containing information about P13, and this is what I found: Therefore, I noticed communication

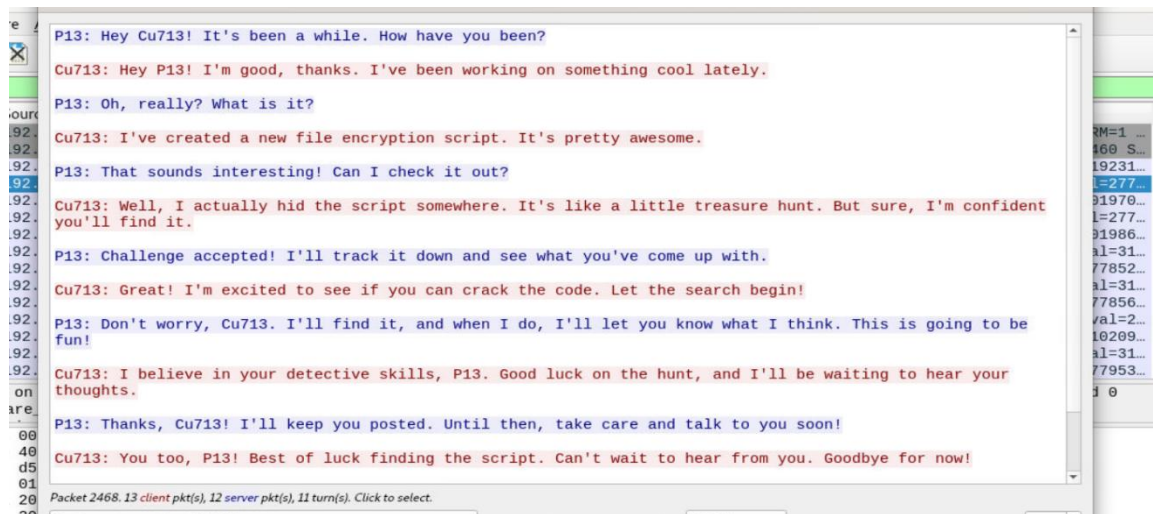


No.	Time	Source	Destination	Protocol	Length	Info
2398	110.139858	192.168.235.137	192.168.235.131	TCP	120	250 → 59576 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=54 TSval=277...
2402	117.034820	192.168.235.131	192.168.235.137	TCP	144	59576 → 250 [PSH, ACK] Seq=1 Ack=56 Win=64256 Len=78 TSval=31...
2410	122.267194	192.168.235.137	192.168.235.131	TCP	95	250 → 59576 [PSH, ACK] Seq=56 Ack=80 Win=65280 Len=29 TSval=2...
2420	134.187582	192.168.235.137	192.168.235.131	TCP	118	250 → 59576 [PSH, ACK] Seq=86 Ack=152 Win=65280 Len=50 TSval=...
2428	145.868702	192.168.235.137	192.168.235.131	TCP	144	250 → 59576 [PSH, ACK] Seq=137 Ack=277 Win=65280 Len=78 TSval...
2452	159.108448	192.168.235.137	192.168.235.131	TCP	177	250 → 59576 [PSH, ACK] Seq=217 Ack=359 Win=65280 Len=111 TSva...
2461	165.084306	192.168.235.131	192.168.235.137	TCP	180	59576 → 250 [PSH, ACK] Seq=360 Ack=328 Win=64256 Len=114 TSva...
2468	170.158044	192.168.235.137	192.168.235.131	TCP	152	250 → 59576 [PSH, ACK] Seq=329 Ack=474 Win=65280 Len=86 TSval...
2481	176.685934	192.168.235.131	192.168.235.137	TCP	166	59576 → 250 [PSH, ACK] Seq=475 Ack=415 Win=64256 Len=100 TSva...
21274	330.208491	194.22.53.86	192.168.235.137	TLSv1.3	1445	Application Data
24883	333.868497	185.80.36.245	192.168.235.137	TCP	1246	443 → 56924 [PSH, ACK] Seq=2905 Ack=518 Win=64240 Len=1192 [T...

Frame 2398: 120 bytes on wire (960 bits), 120 bytes captured (960 bits) on interface \Device\NPF{1CBF945E-5CD1-44C7-A837-D87411311052}, id 0
Ethernet II, Src: VMware_55:9d:d4 (00:0c:29:55:9d:d4), Dst: VMware_39:70:04 (00:0c:29:39:70:04)

```
0000 00 0c 29 39 70 04 00 0c 29 55 9d d4 00 00 45 00  -->P... )U...E...
0010 00 6a 1e a9 40 00 00 06 c3 86 c0 a8 eb 89 c0 a8  --j...@...
0020 eb 83 00 fa e8 b8 d5 71 8b 63 51 5b 22 f8 80 18  --.....q..cQ["...
0030 01 fe d5 bb 00 00 01 01 00 0a 10 0e 8f c1 b9 61  --.....a
0040 c4 bf 50 31 33 3a 20 48 65 79 20 43 75 37 31 33  --P13: H ey Cu713
0050 21 20 49 74 27 73 20 62 65 65 6e 20 61 20 77 68  --! It's b een a wh
0060 69 6c 65 2e 20 48 6f 77 20 68 61 76 65 20 79 6f  --ile. How have yo
0070 75 20 62 65 65 6e 3f 0a                          --u been?..
```

between two IP addresses and determined that the sender was **192.168.235.137**, which is **P13**, and the receiver was **192.168.235.131**, which is **Cu713**.



No.	Time	Source	Destination	Protocol	Length	Info
2468	170.158044	192.168.235.137	192.168.235.131	TCP	152	250 → 59576 [PSH, ACK] Seq=329 Ack=474 Win=65280 Len=86 TSval...

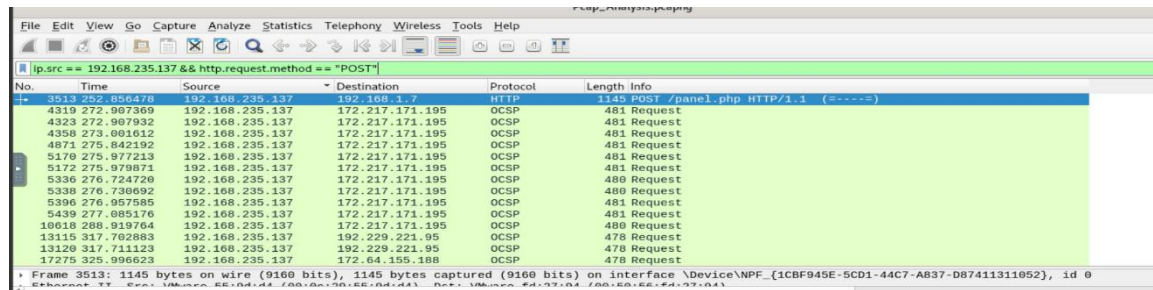
Packet 2468. 13 client pkt(s), 12 server pkt(s), 11 turn(s). Click to select.

```
P13: Hey Cu713! It's been a while. How have you been?
Cu713: Hey P13! I'm good, thanks. I've been working on something cool lately.
P13: Oh, really? What is it?
Cu713: I've created a new file encryption script. It's pretty awesome.
P13: That sounds interesting! Can I check it out?
Cu713: Well, I actually hid the script somewhere. It's like a little treasure hunt. But sure, I'm confident you'll find it.
P13: Challenge accepted! I'll track it down and see what you've come up with.
Cu713: Great! I'm excited to see if you can crack the code. Let the search begin!
P13: Don't worry, Cu713. I'll find it, and when I do, I'll let you know what I think. This is going to be fun!
Cu713: I believe in your detective skills, P13. Good luck on the hunt, and I'll be waiting to hear your thoughts.
P13: Thanks, Cu713! I'll keep you posted. Until then, take care and talk to you soon!
Cu713: You too, P13! Best of luck finding the script. Can't wait to hear from you. Goodbye for now!
```

Q2: P13 uploaded a file to the web server. What is the IP address of the server?

This question asked me to find the IP address of the server that uploaded the file, so I focused on POST because that's where the upload was done:

ip.src == 192.168.235.137 && http.request.method == "POST"

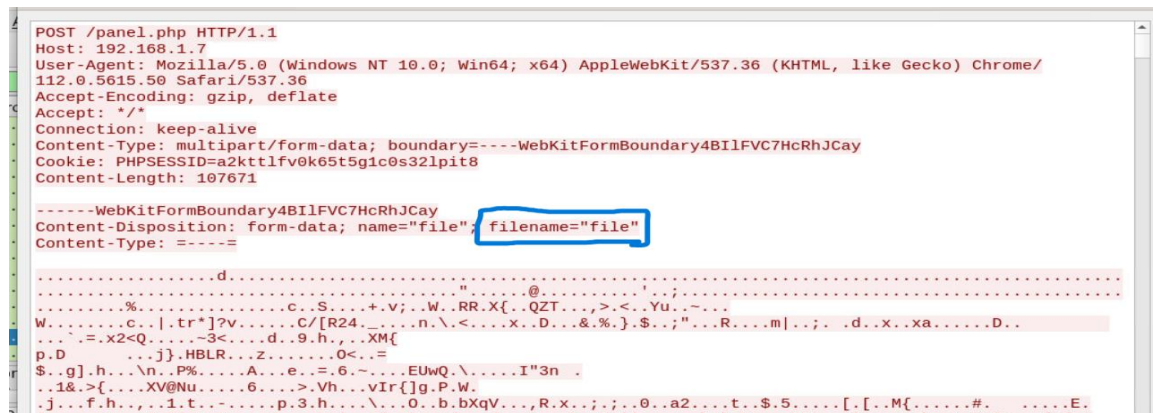


No.	Time	Source	Destination	Protocol	Length	Info
3513	252.856478	192.168.235.137	192.168.1.7	HTTP	1145	POST /panel.php HTTP/1.1 (=====)
4319	272.907369	192.168.235.137	172.217.171.195	OCSP	481	Request
4323	272.907932	192.168.235.137	172.217.171.195	OCSP	481	Request
4358	273.001612	192.168.235.137	172.217.171.195	OCSP	481	Request
4871	275.842192	192.168.235.137	172.217.171.195	OCSP	481	Request
5170	275.977213	192.168.235.137	172.217.171.195	OCSP	481	Request
5172	275.979871	192.168.235.137	172.217.171.195	OCSP	481	Request
5336	276.724720	192.168.235.137	172.217.171.195	OCSP	480	Request
5338	276.730692	192.168.235.137	172.217.171.195	OCSP	480	Request
5396	276.957585	192.168.235.137	172.217.171.195	OCSP	481	Request
5430	277.005176	192.168.235.137	172.217.171.195	OCSP	481	Request
10618	288.919764	192.168.235.137	172.217.171.195	OCSP	480	Request
13115	317.702883	192.168.235.137	192.229.221.95	OCSP	478	Request
13120	317.711123	192.168.235.137	192.229.221.95	OCSP	478	Request
17275	325.996623	192.168.235.137	172.64.155.188	OCSP	478	Request

So, ip server: 192.168.1.7

Q3: What is the name of the file that was sent through the network?

I checked the packet to find the filename and found it: its name is **file**

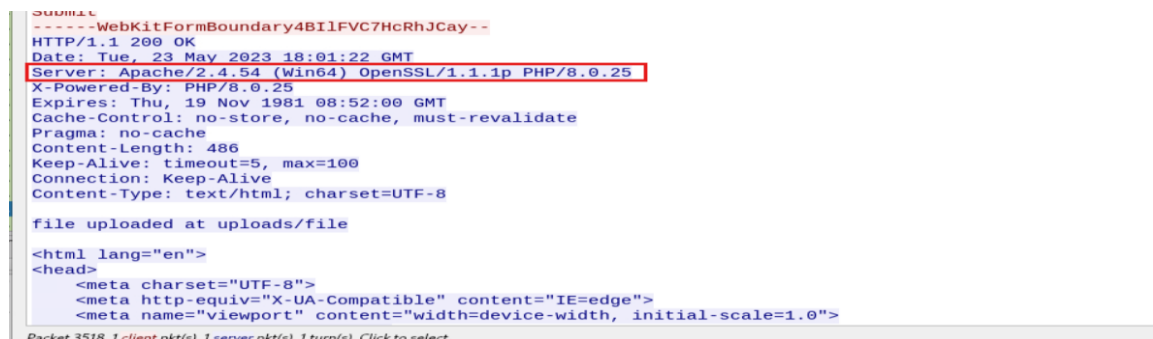


```
POST /panel.php HTTP/1.1
Host: 192.168.1.7
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.50 Safari/537.36
Accept-Encoding: gzip, deflate
Accept: */*
Connection: keep-alive
Content-Type: multipart/form-data; boundary=---WebKitFormBoundary4BILFVC7HcRhJCay
Cookie: PHPSESSID=a2kttlfv0k65t5g1c0s32lpit8
Content-Length: 107671

-----WebKitFormBoundary4BILFVC7HcRhJCay
Content-Disposition: form-data; name="file"; filename="file"
Content-Type: -----
```

Q4: What is the name of the web server where the file was uploaded?

In the same packet, I found the server name, and it was Apache.



```
HTTP/1.1 200 OK
Date: Tue, 23 May 2023 18:01:22 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.0.25
X-Powered-By: PHP/8.0.25
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Content-Length: 486
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=UTF-8

file uploaded at uploads/file

<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

Q5: What directory was the file uploaded to?

Also, in the same packet, I was able to find the directory and its name: **Uploads**



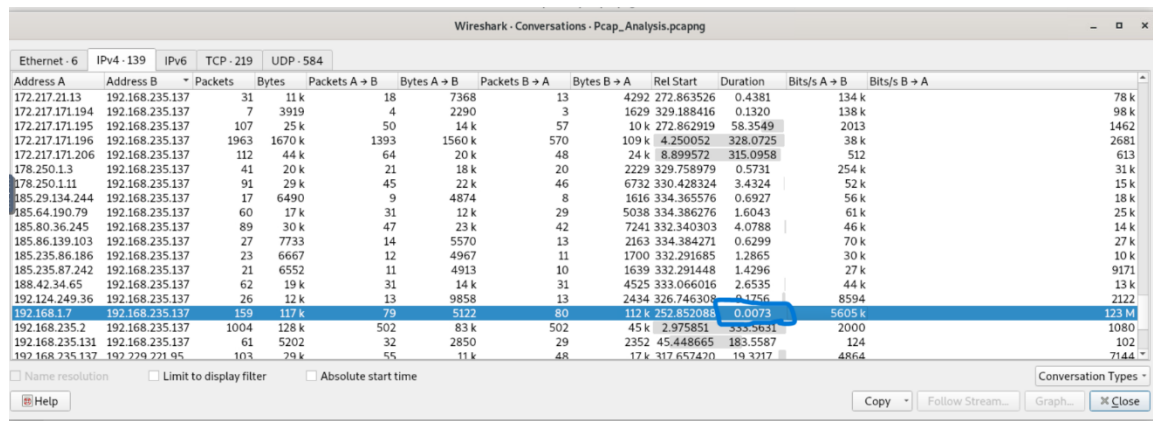
```
Submit
-----WebKitFormBoundary4BilFVC7HcRhJCay--
HTTP/1.1 200 OK
Date: Tue, 23 May 2023 18:01:22 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1p PHP/8.0.25
X-Powered-By: PHP/8.0.25
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Content-Length: 486
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=UTF-8

file uploaded at uploads/file

<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
</body>
</html>
```

Q6: How long did it take the sender to send the encrypted file?

From the question, we note that it was sent from the server to the user 192.168.235.137. This was confirmed because the file data was present within the HTTP (200 OK) response issued from the server's IP address. To determine the transmission time take sender to send the encrypted file, I analyzed the 'Conversations' statistics. I specifically looked for the stream originating from the server (192.168.1.7) to the user (192.168.235.137). The 'Duration' field confirmed the exact time it took to send the file.



Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
172.217.21.13	192.168.235.137	31	11 k	18	7368	13	4292	272.863526	0.4381	134 k	78 k
172.217.171.194	192.168.235.137	7	3919	4	2290	3	1629	329.188416	0.1320	138 k	98 k
172.217.171.195	192.168.235.137	107	25 k	50	14 k	57	10 k	272.862919	58.3549	2013	1462
172.217.171.196	192.168.235.137	1963	1670 k	1393	1560 k	570	109 k	4.250052	328.0725	38 k	2681
172.217.171.206	192.168.235.137	112	44 k	64	20 k	48	24 k	8.899572	315.0958	512	613
178.250.1.3	192.168.235.137	41	20 k	21	18 k	20	2229	329.758979	0.5731	254 k	31 k
178.250.1.11	192.168.235.137	91	29 k	45	22 k	46	6732	330.428324	3.4324	52 k	15 k
185.29.134.244	192.168.235.137	17	6490	9	4874	8	1616	334.365576	0.6927	56 k	18 k
185.64.190.79	192.168.235.137	60	17 k	31	12 k	29	5038	334.386276	1.6043	61 k	25 k
185.80.36.245	192.168.235.137	89	30 k	47	23 k	42	7241	332.340303	4.0788	46 k	14 k
185.86.139.103	192.168.235.137	27	7733	14	5570	13	2163	334.384271	0.6299	70 k	27 k
185.235.86.186	192.168.235.137	23	6667	12	4967	11	1700	332.291685	1.2865	30 k	10 k
185.235.87.242	192.168.235.137	21	6552	11	4913	10	1639	332.291448	1.4296	27 k	9171
188.42.34.65	192.168.235.137	62	19 k	31	14 k	31	4525	333.066016	2.6535	44 k	13 k
192.124.249.36	192.168.235.137	26	12 k	13	9858	13	2434	326.746308	0.4756	8594	2122
192.168.1.7	192.168.235.137	159	117 k	79	5172	80	112 k	252.652088	0.0073	5605 k	123 k
192.168.235.2	192.168.235.137	1004	128 k	502	83 k	502	45 k	2.975851	333.5631	2000	1080
192.168.235.131	192.168.235.137	61	5202	32	2850	29	2352	45.448665	183.5587	124	102
192.168.235.137	192.229.221.95	103	29 k	55	11 k	48	17 k	317.657420	19.3217	4864	7144

Summary

An encryption script named 'file' was discovered on the local Apache web server after being uploaded via '/panel.php' to the '/uploads' directory during a challenge between users P13 and Cu713.